Momentum & Contrarian Strategies on UK Assets — Extension Study

A reproducible re-examination of an undergraduate dissertation, extended to July 2026. Synthesis of Phases 1-5, with Phase 6 (stock-level generalisation) and Phase 7 (portfolio-construction) addenda; every figure is traceable to a committed output in `reports/`.

Executive summary

This project rebuilt and extended a dissertation testing seven momentum and contrarian trading strategies on the FTSE 100 (via the VUKE.L ETF) and GBP/USD over 2015-2025. Four findings stand out. **First, momentum failed structurally on the FTSE 100** — all four momentum strategies lost money, three significantly underperforming a passive benchmark. **Second, the one strategy that worked in-sample was the Contrarian Bollinger Band** — 7.6% CAGR at a 0.69 Sharpe while invested only ~44% of the time — and I reproduced that independently, within a pre-registered tolerance, on a fresh July 2026 data pull (~11 months after the original sample ended). **Third, that edge did *not* persist out-of-sample:** over the ~11-month extension it lost 1.5% while the market rose 21.6%. **Fourth, what survives scrutiny is narrow but real:** the edge is robust to its parameters, and its high-volatility regime premium holds up (attenuated) once a look-ahead bias in the original regime labels is removed. The honest conclusion is an edge that has, at best, narrowed and did not survive the extension — though whether that reflects genuine decay or simply an unfavourable regime cannot be resolved over ~11 months. Either way, it is not a deployable edge as it stands.

What was built

A clean, reproducible Python system: a snapshot-driven data pipeline (results reproduce from committed CSVs, not live API calls), the seven strategies **re-implemented independently from the dissertation's written methodology** (not the original code), a shared next-day-execution backtest engine, regime and robustness analysis, and a minimal live-signal harness. The work proceeded in gated phases, each with a plan committed before the build and an honest report committed after.

Original results, validated

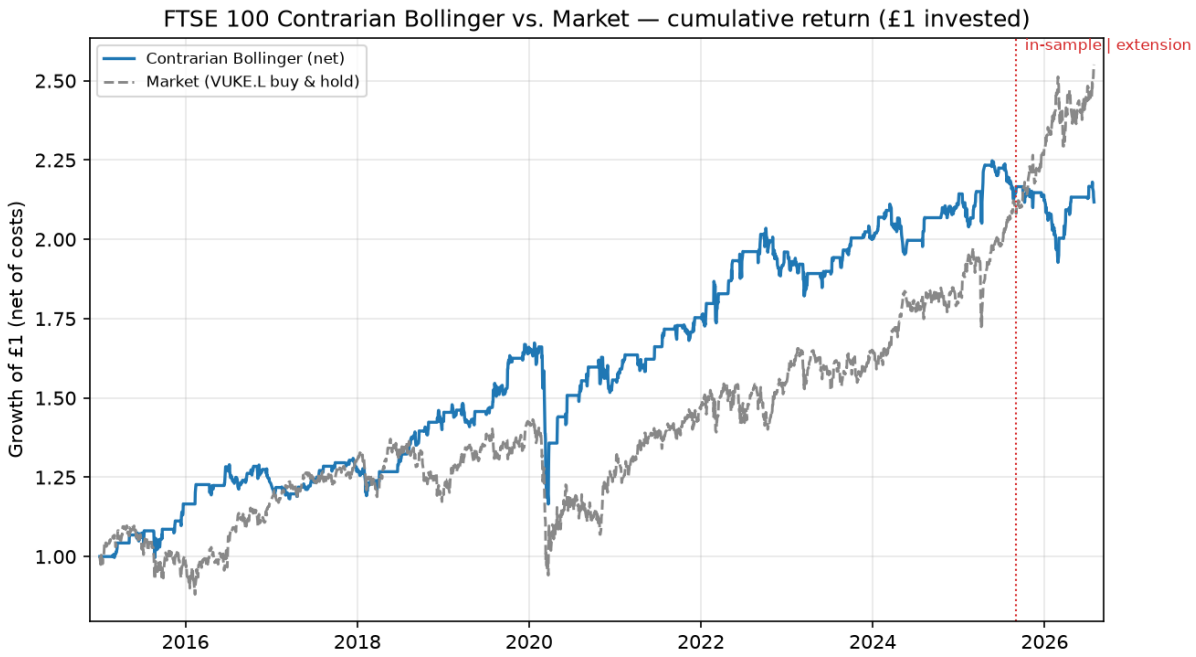
Re-implementing from the paper and running on a fresh 2015-Aug 2025 pull reproduced every headline figure within tolerance (CAGR ±1pp, Sharpe ±0.1, HAC t ±0.3). Selected FTSE 100 results (mine / dissertation):

Strategy	CAGR	Sharpe	Note
Market (buy & hold)	7.2% / 7.2%	0.46 / 0.46	anchor matched exactly
Contrarian Bollinger	**7.4% / 7.6%**	**0.67 / 0.69**	only strategy to beat/match the market
Momentum EMA (worst)	–12.0% / –11.8%	–0.77 / –0.76	–19.5% ann. differential, HAC-significant

All four momentum strategies produced negative CAGRs; three (SMA, EMA, Lookback) significantly underperformed the market under HAC-robust testing. GBP/USD showed no persistent edge for any

strategy. The re-implementation deltas ran uniformly *slightly worse* than the original (consistent with vendor data revisions, and the opposite direction to what look-ahead bias would cause).

Extension to July 2026 — did the edge persist?



The chart shows the story at a glance: the Contrarian Bollinger (blue) leads the market through the in-sample period, then the market pulls decisively ahead after the Aug-2025 split. Over the extension window (Sept 2025 -> Jul 2026, ~230 trading days) the strategy returned –**1.5% cumulative versus the market's +21.6%.**

> **Directional only.** ~11 months (~100 active days) is far too short for statistical significance; no significance is claimed on this window. The verdict is "did not persist," not a quantified decay rate.

Why performance changed

The extension result is consistent with the dissertation's central thesis: a behavioural, sentiment-driven edge that erodes as the UK equity market's structure shifts toward passive ownership, compressing the return autocorrelation that mean-reversion rules exploit. The strongest single piece of original evidence for that thesis was a **monotonic decay in the strategy's Fama-French five-factor alpha, from 10.4% (pre-Brexit) to 1.9% (inflation era).**

> **Attribution:** these decay figures (10.4% -> 1.9%) are the **original dissertation's sub-period finding and were *not* re-derived in this project** — sub-period factor attribution was outside the re-implementation scope. They are cited here as context, not as an independent reproduction. What *this* project independently establishes is the out-of-sample non-persistence above and the regime results below.

Robustness and the look-ahead fix

Two credibility checks matter most for anyone assessing whether the in-sample edge was real or curve-fit:

- **Parameter robustness.** The Bollinger edge is positive across almost the entire band-window {10-40} x width {1.5-2.5sigma} grid — it does not depend on the exact (20, 2sigma) choice. The two regime-classification parameters flagged by the dissertation's markers were also varied: the high-volatility premium stays positive across vol-state windows {10, 20, 40 days} (+7.8 / +5.6 / +4.8pp) and the low-rate outperformance stays positive across Bank Rate thresholds {0.5%, 1%, 2%} (+7.0 / +4.8 / +7.2pp). The finding does not hinge on those choices.
- **Removing a look-ahead bias.** The original labelled volatility regimes using the *full sample* — information a real-time trader would not have had. Re-estimating with a walk-forward Markov model (parameters fit only on the expanding past window; one-sided filtered probabilities) attenuates the high-volatility premium from the original **15.8% to 13.1%** (vs 6.1% in low-vol) — smaller, but still clearly present. The 13.1% is the honest number; the 15.8% used information unavailable in real time.

Limitations — what decayed, why, and what I'd test next

This section is deliberately the most developed part of the report: the conclusions are only as good as the caveats attached to them.

The most damaging limitation, stated plainly: the extension cannot distinguish decay from an unfavourable regime. A contrarian, mean-reversion strategy that is in cash ~56% of the time (and net-long only a minority of days) will *structurally* lag a strong, one-directional bull market — and the extension window was exactly that (the market rose 21.6%). So the strategy's –1.5% is as consistent with "contrarian strategies trail in a rally" as it is with "the edge has decayed." The "Why performance changed" section above should be read with this in mind: the extension is *consistent with* the decay thesis but does not, on its own, establish it. *What would actually resolve it — the clear next step, not built here:* a **regime-matched out-of-sample comparison**, measuring how the strategy performs specifically in *trending* versus *choppy / high-volatility* out-of-sample stretches, rather than over a single calendar window dominated by one regime. That is the test that would separate genuine decay from regime drag.

The in-sample edge is subject to multiple-testing / selection bias. Seven strategies across two assets and several regime cuts is a large hypothesis family, and the Contrarian Bollinger was in part *selected because it worked*. The parameter-robustness grid guards against knife-edge overfitting, but it is itself in-sample, so it does not correct for the strategy having been chosen ex-post. Harvey, Liu and Zhu (2016) show that isolated "significant" results within large families are frequently chance. *Next step:* a **deflated Sharpe ratio or bootstrap** that explicitly penalises for the number of strategies and parameter combinations tried.

The extension window is short (~11 months, ~100 active days). It supports a directional read only; no significance is claimed. *Next step:* re-run periodically as more out-of-sample data accrues.

Single-ETF proxy and a lower-bound cost model. VUKE.L carries tracking error versus the index itself, and the 0.04% round-trip cost model excludes CFD financing, market impact, and elevated short-selling costs during stress — so realised costs would be higher, especially on the short leg. *Next step:* re-price under a stress-dependent cost model.

Low factor explanatory power. FF5 R2 is ~0.15 — standard factors explain little of these daily strategy returns (partly a currency mismatch: European factors in USD vs GBP returns), so factor attribution can neither confirm nor rule out an omitted-risk explanation for the alpha.

Deferred by design. The event studies (FOMC/Trump/BoE) were not re-run; the markers themselves flagged that daily-frequency event windows are ill-identified, so this is a known, deliberate gap rather than

an oversight.

Reference-data caveat. The Bank Rate series is a manually-entered static snapshot of the official BoE schedule; the threshold-driving values are the well-established ones, but it should be spot-checked before being quoted precisely.

The live signal

A minimal harness (`live_signal.py`) generates the current Bollinger signal and logs it. It is a **capability demonstration only** — no broker, no capital, no orders — and, given the non-persistence finding above, explicitly **not** a system I would trade.

Stock-level generalisation check (Phase 6)

A generalisation check (`reports/phase6_stock_generalisation.md`) applied the exact validated Contrarian Bollinger strategy to eight individual FTSE 100 constituents, one per sector, over the original 2015-Aug 2025 window — a **mechanism check, not a signal search**. The result is an honest null: five of eight names showed a positive strategy CAGR and three beat their own buy-and-hold, but **none was HAC-significant even at a raw α=0.05, and none survived a Bonferroni correction** ($\alpha/8 \approx 0.00625$) for the eight parallel tests. This is entirely consistent with the index-level non-persistence above: the edge does not generalise to single names, reinforcing that the in-sample result was an artefact rather than a robust, tradable phenomenon.

Portfolio construction — a capability demonstration (Phase 7)

A later extension (`reports/phase7_portfolio.md`) steps up from single strategies to the portfolio level: an inverse-volatility-weighted portfolio of eight single-name Contrarian Bollinger sleeves, reported with portfolio-level risk metrics (return, volatility, Sharpe, correlation structure, drawdown). It is a **construction-capability demonstration, not a tradable portfolio** — it is built on the same Contrarian Bollinger signal shown above to fail out-of-sample and (across eight individual stocks, Phase 6) to lack any statistically significant edge, so every figure is in-sample and inherits that null result. In-sample the inverse-vol portfolio shows a higher Sharpe (0.54 vs 0.30) and a shallower drawdown (–18.8% vs –27.5%) than an equal-weight buy-and-hold of the same names — but that improvement is **mechanical, not alpha**: the portfolio sits in cash ~48% of the time and deliberately sizes down the most volatile names, and its return is statistically indistinguishable from the benchmark (HAC p = 0.89). The point is to demonstrate portfolio sizing, correlation, and drawdown assessment — including that the in-sample correlations are unstable (mean pairwise 0.21 -> 0.04 across the two halves of the sample) — not to propose anything to trade.

Conclusion

Momentum is structurally broken on the FTSE 100 in the passive-investing era. The Contrarian Bollinger Band captured a genuine behavioural premium in-sample — robust to its parameters and concentrated in high-volatility, low-rate conditions — but that premium has, at best, narrowed: it did not survive the out-of-sample extension, and it shrinks once a look-ahead bias is removed. Whether the out-of-sample failure reflects genuine decay or simply an unfavourable regime cannot be settled over ~11 months, and the report is explicit about that. The defensible takeaway is not "here is a strategy that beats the market," but "here is a once-promising edge, honestly measured, that no longer clears the bar — with the open question of why stated plainly."